Network Topology and Architecture

3

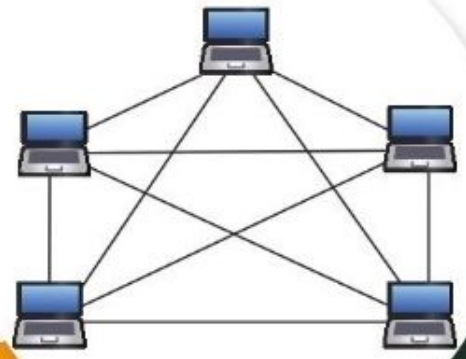
A network comprises all the hardware and software components involved in connecting computers across small and large distances. An internetwork is a collection of independent networks connected by different networking devices.

Visual representation is very helpful while displaying complex network information such as all the devices included in the internetwork and the medium of a large internetwork. A figure, using symbols to represent the different devices and connections of an internetwork, provides an easy way to understand such complex information. This type of diagram of a network is known as a topology diagram. This chapter will deal with the various network topologies and their advantages and disadvantages.

OBJECTIVES

By the end of this chapter, you will be able to:

- Describe various network topologies, their advantages, and disadvantages
- Describe the client-server and peer-to-peer architecture
- Understand the Cisco three-layer hierarchical model



A network can be of different sizes ranging from a simple network to a network connecting millions of devices. A network architecture defines the role of each computer and the way it is organised in the network. This chapter provides information about the different types of network architectures.

Network Topology

A topology refers to the way in which computers are connected to each other. It defines the layout, virtual shape or structure of the network. A topology can either be physical or logical. Physical topology refers to the physical layout of the network components and cables and a logical structure defines the flow of information between the computers.

CERTIFICATION OBJECTIVE

Describe various network topologies, their advantages, and disadvantages

The four basic network topologies are: mesh, star, bus, and ring.

Bus Topology

Bus topology is the simplest topology. In this topology, all the nodes, computers, and servers are connected to a single cable called the bus. This single long cable acts as the backbone and all the nodes are connected to it by droplines and taps. The connection between the main cable and device is known as dropline. The connector, which creates the contact with the metallic core of the main cable, is known as a tap. When the signal travels through the backbone, some of its energy is transformed into heat. The farther the signal travels, the weaker it becomes. Figure 1 shows an example of bus topology.

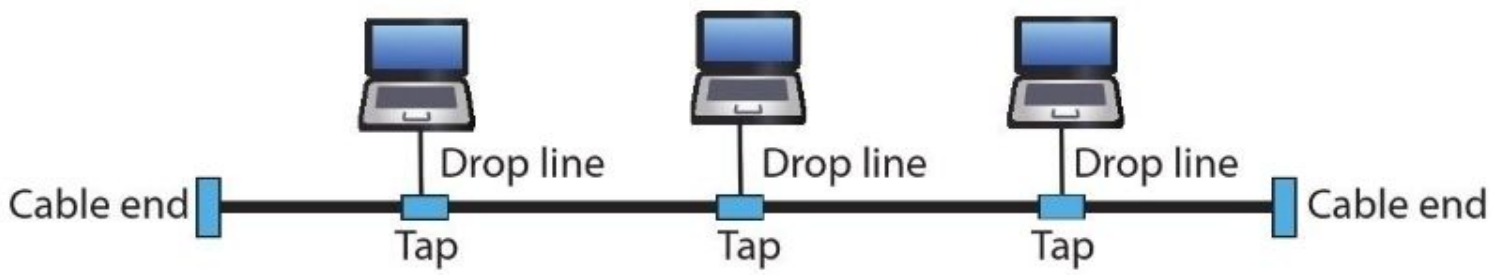


Figure 1: Bus topology

To prevent bouncing of signals, a terminator is added to the backbone. A barrel connector is used to extend the cable. When the source broadcasts a signal, it travels to all the workstations connected by the cable. Only the machine whose MAC or IP address is matched accepts the signal while all the other machines reject it.

Some of the advantages and disadvantages of using a bus topology are listed below:

Advantages

- Easy to install.
- Requires less cabling when compared to other topologies.
- Costs less.
- Works well for small networks.

Disadvantages:

- The entire network shuts down in case of any fault in the network.
- Difficult to troubleshoot if the network shuts down.
- Difficult to add new devices.
- Signal reflection at taps degrade the signal quality.
- Not suitable for busy networks as the packet collision rate is very high.
- All the computers in the network receive the signal, therefore, the security is low.

Ring Topology

A ring topology has a dedicated point-to-point connection with only two devices on either side of it. All the nodes are connected to each other in such a way that they form a closed loop. Each node is connected to two nodes on either side and it communicates with only its adjacent neighbours. The data travels in the network in one direction. A signal travels to every node in one direction until it reaches its

destination. Every device has a repeater. Whenever a signal passes through the node, it regenerates the signal and sends it. Figure 2 shows an example of ring topology.

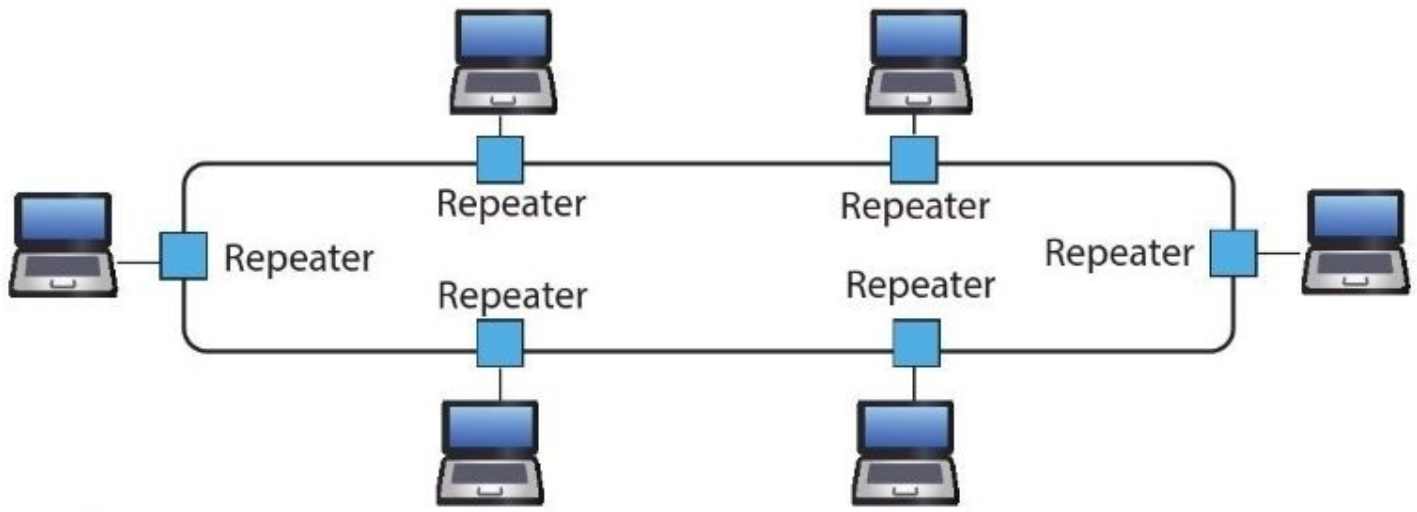


Figure 2: Ring Topology

Some of the advantages and disadvantages of using a ring topology are listed below:

Advantages:

- Easy to install and reconfigure - Adding or deleting a device requires changing only two connections.
- Performs better than the bus topology under heavy network load.
- Fault isolation is easy because of point-to-point connection.
- No network server required to control the connectivity between nodes.

Disadvantages:

- In a unidirectional ring, a break can disable the entire network.
- Slower than star topology because each packet of data should pass through all the nodes between the source and the destination.

Star Topology

A star topology has dedicated point-to-point links between the devices and the central controller, usually a hub. The devices are not linked to each other. If a device wants to send the data to one device, it sends the data to the controller, which then passes the data to the destination device. All data must pass through the central controller before reaching the destination.

The hub acts as the connector for different nodes. It manages and controls the whole network. Figure 3 illustrates star topology.

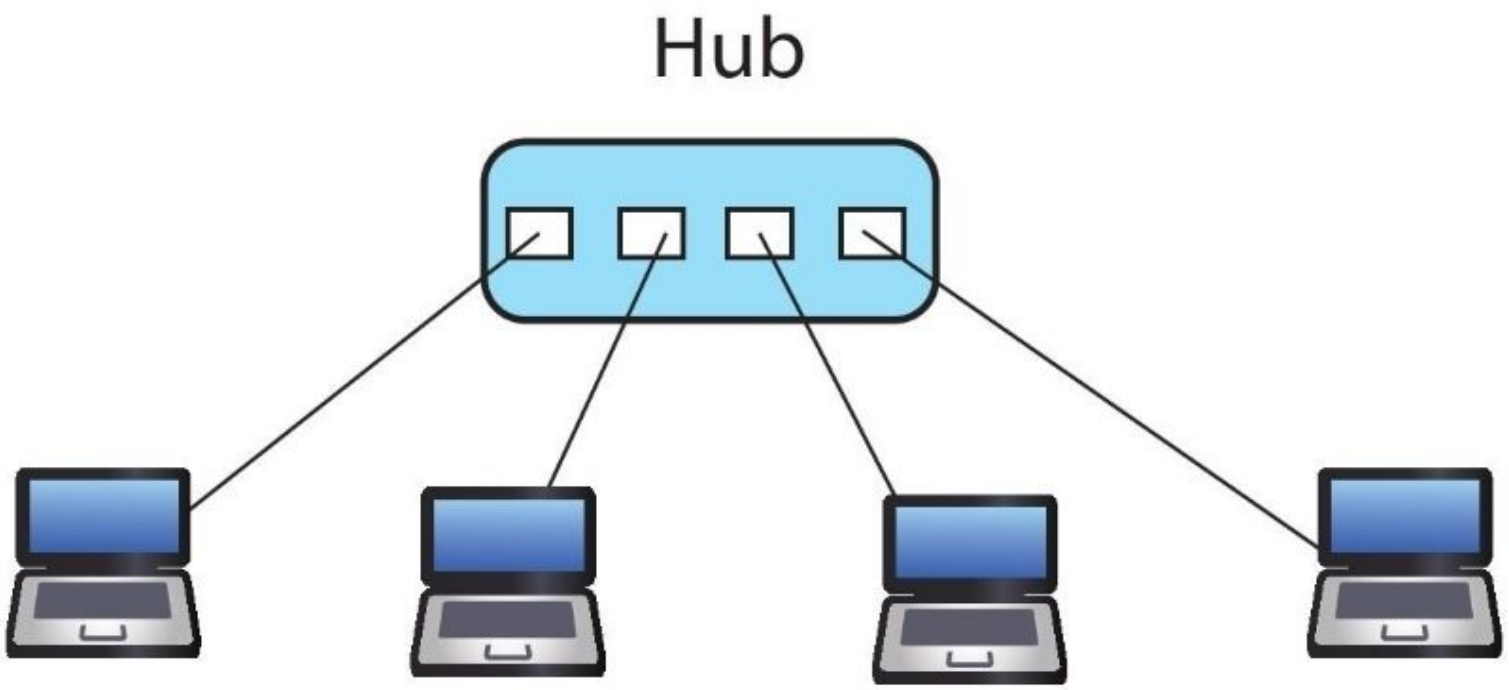


Figure 3: Star topology

Advantages:

- Less expensive compared to mesh topology.
- Each device needs only one link and one I/O port to connect to any number of nodes. This makes installation and reconfiguration easier.
- Robust in nature -- if one link fails, the other links are not affected.
- Easy fault isolation and identification mechanism.
- Centralised management.
- Signal does not transmit to all nodes, therefore, it has better performance than the bus topology.

Disadvantages:

- The whole topology depends on the hub, which is the central controller. So, if the hub goes down, the whole system goes down.
- Requires more cabling than the bus or star topology.
- Performance depends on the central device.
- More expensive because of the central device used.

Mesh Topology

A mesh topology has a dedicated point-to-point link between every device. Every node in the mesh topology sends its own data but also relays data from other nodes. It uses a flooding or routing technique. Every node cooperates in the distribution of data in the network. Figure 4 shows an example of mesh topology.

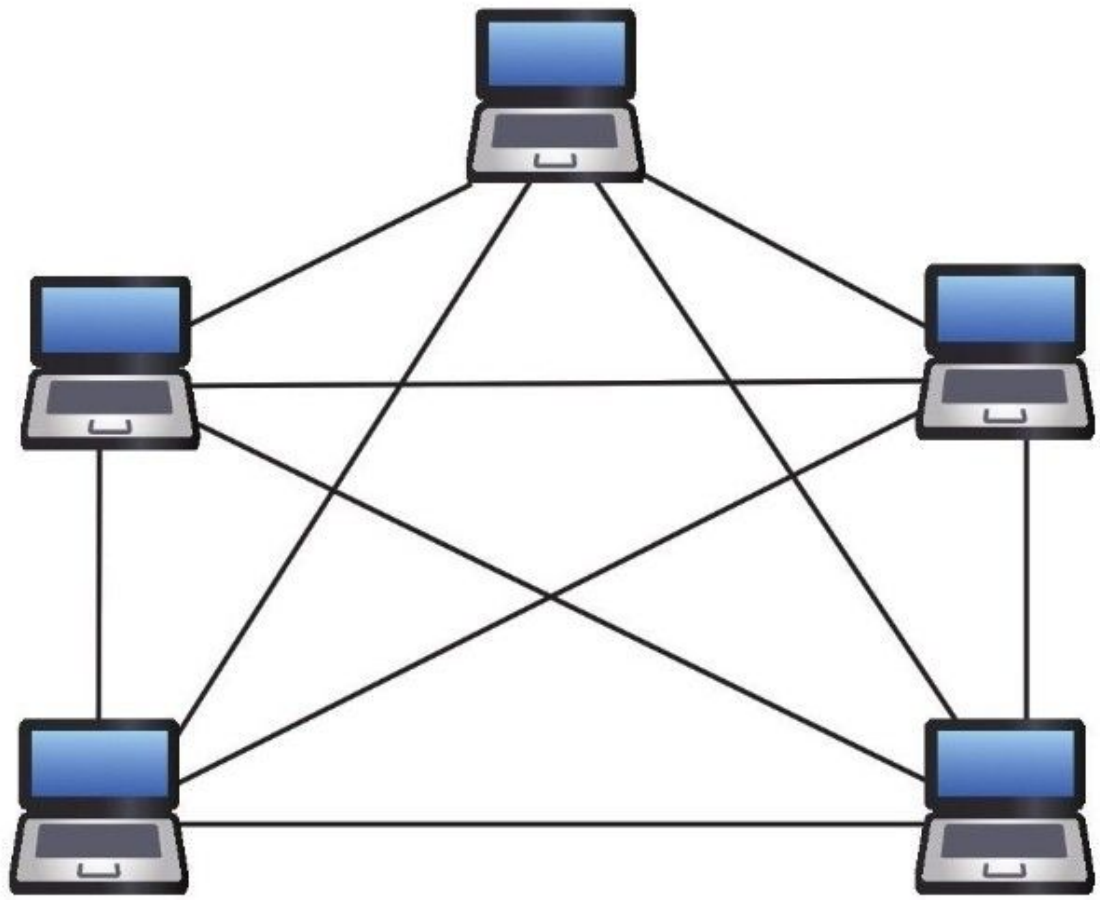


Figure 4: Mesh topology

Mesh topology can be of two types – full and partial.

In a full mesh topology, every node is linked to every other node in the network. It is very expensive to implement and is redundant. Its main advantage is that if any link fails, then data can be passed through any other link. It is used in the backbone network.

IMPORTANT NOTES

In full mesh topology, if there are n nodes, then it requires n(n-1)/2 full-duplex nodes to create a full mesh.

In a partial mesh topology, some nodes are arranged as a full mesh but some nodes are connected to only one or two nodes. It is less expensive and less redundant.

Advantages:

- Eliminates traffic problems by using dedicated links between devices.
- Robust in nature -- Failure of one link does not affect the other links.

- Secure and private -- Only the intended recipient will receive the data packet.
- Point-to-point links make fault isolation and identification easy.
- Expansion and modification is possible without disturbing other nodes.

Disadvantages:

- Large amount of cabling and I/O ports are required.
- Difficult to install and reconnect.
- Overall cost of this topology is very high compared to others.

ANSWER IT

1. In a fully connected mesh topology with seven computers, we need _____ links.
   a. 7 c. 21
   b. 42 d. None of the choices are correct
2. Which topology contains a backbone which runs the length of the network?
   a. Star c. Bus
   b. Tree d. Ring
3. Which topology requires a central controller or hub?
   a. Mesh c. Bus
   b. Star d. Ring

Network Architecture

A network architecture defines how to connect computers and the roles they perform. It defines the design and framework of the network. It also explains the characteristics of individual hardware, software, and transmission system components. Depending on their architecture, computer networks are divided into two -- client-server and peer-to-peer.

CERTIFICATION OBJECTIVE

Describe the client-server and peer-to-peer architecture

Client-Server

In a client-server architecture, there are client and server nodes. Servers are nodes that have software which enable them to provide services and information. Each service requires a separate software. For example, a web server software is required to provide web services. Clients are nodes that have software installed on the system to request services of the server. An example of the client software is a

web browser, like the Internet Explorer. Figure 5 shows an example of client-server architecture.

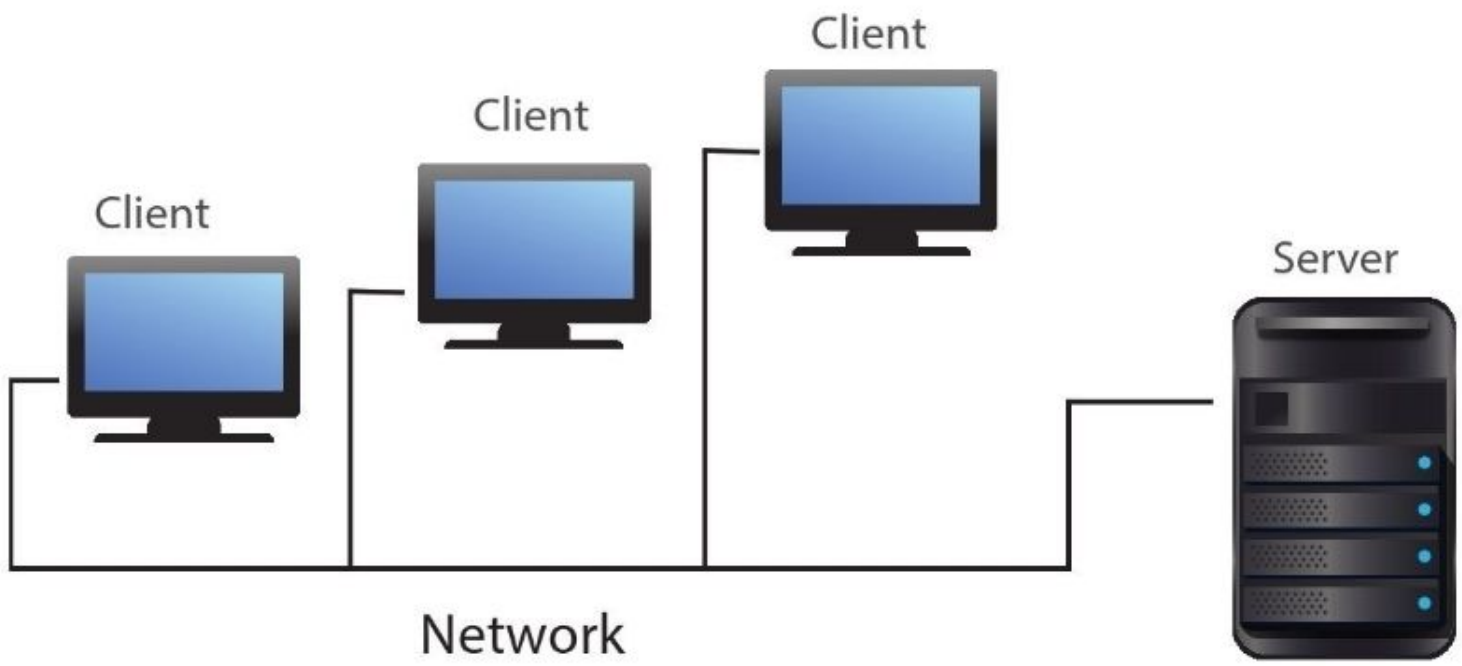


Figure 5: Client-server architecture

A server can provide many services simultaneously to one or many clients. Like a server, a client can also request many services from different servers. For example, a user can check an email and view a web page together. There is a 1:M relationship between a server and its clients.

Peer-to-Peer

In the client-server architecture, the client and the server usually run on different nodes. But, it is possible for a node to carry out both the functions of the client and the server at the same time. When a node functions as a client and server simultaneously, then this type of network is called a peer-to-peer network.

A simple peer-to-peer network can have two directly connected nodes using a wired or wireless connection. Multiple nodes connected to a network device, such as a hub, create a larger peer-to-peer network. One disadvantage of the peer-to-peer architecture is that the performance of a node can slow down because it acts both as a client and a server at the same time.

Figure 6 shows an example of peer-to-peer architecture.

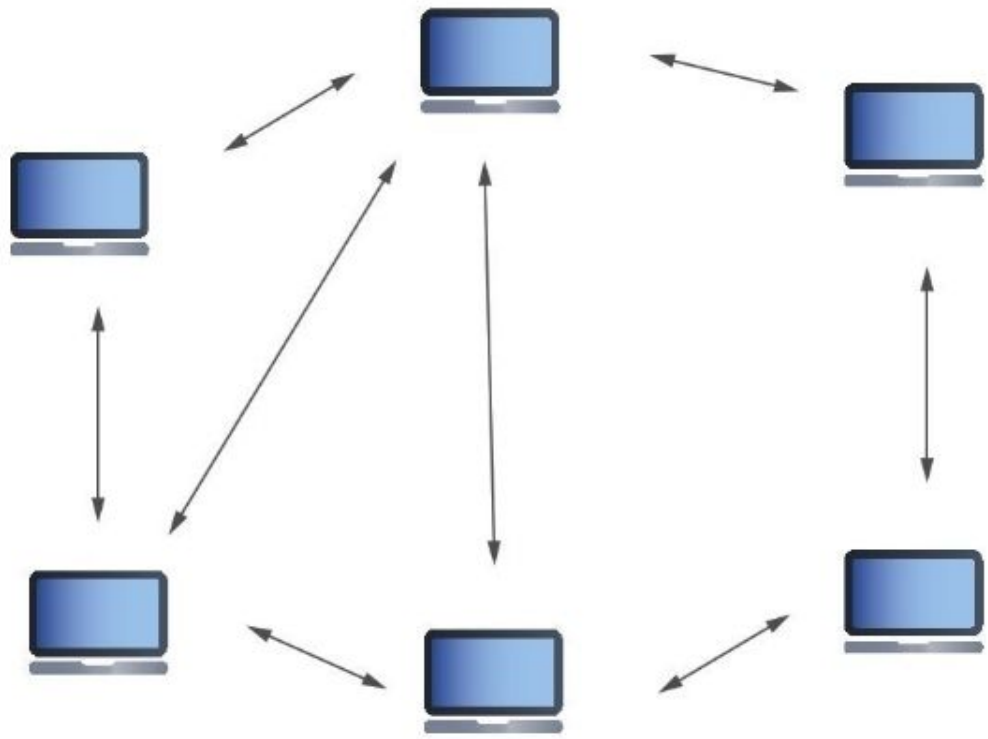


Figure 6: Peer-to-peer architecture

Table 1 draws the comparison of peer-to-peer architecture and client-server architecture.

Table 1. Comparison between peer-to-peer and client-server architecture

Peer-to-peer architecture	Client-server architecture
Easy to set up.	More difficult to set up.
Less expensive to install.	More expensive to install.
Can be implemented on a wide range of operating systems.	Client can be implemented on various operating systems, but server needs an operating system that supports networking.
Maintaining the software is very time consuming because every node is to be maintained individually.	Maintaining the software is less time consuming because most of the maintenance is done on the server side.
Low level of security.	High level of security.
Ideal choice for a network of less than ten nodes.	There is no limit to the computers that are supported by the server.

The Cisco Three-Layer Hierarchical Model

CERTIFICATION OBJECTIVE

Understand the Cisco three-layer hierarchical model

Networks that have multiple protocols, detailed configurations, and different technologies are very hard to design, implement and maintain. The Cisco hierarchical model helps in designing, implementing, and maintaining a scalable, reliable, and cost-effective hierarchical internetwork. In the Cisco hierarchical model, there are three layers of hierarchy which are shown in Figure 7. These three layers and their typical functions are:

- The core layer - backbone
- The distribution layer - routing
- The access layer - switching

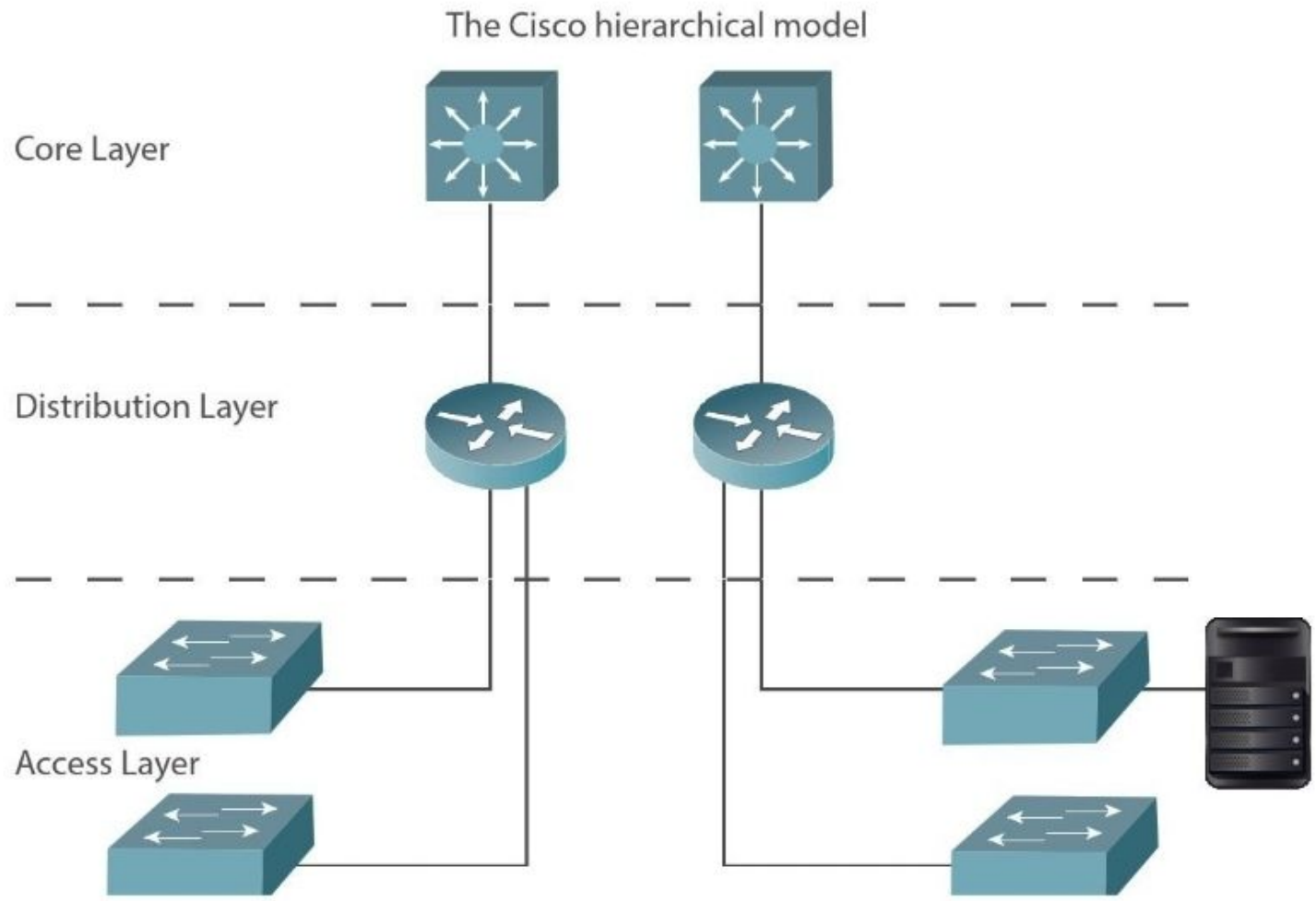


Figure 7: Cisco three-layer hierarchical model

The Core Layer

The core layer is the top layer of the hierarchy and the core of the network. It transports large amounts of traffic quickly and reliably. The function of this layer is to switch traffic as fast as possible. If there is any failure in the core, then every single user can be affected. Therefore, fault tolerance, speed, and latency are important issues in this layer.

The Distribution Layer

The distribution layer enables the communication between the access layer and the core layer. The distribution layer performs routing, filtering, and determines how the packets can access the core. It determines the fastest way to service the network requests. Once this layer decides the route, it forwards the request to the core layer. Then, the core layer transports the request to the correct service. Policies are implemented in the distribution layer. The actions performed at the distribution layer are:

- Routing
- Implementing packet filtering and queuing
- Implementing security and network policies
- Redistributing between routing protocols, including static routing
- Defining broadcast and multicast domains

The Access Layer

The access layer is also known as the desktop layer. It controls the access of users and workgroups to internetwork resources. The access layer is sometimes referred to as the desktop layer. In this layer, the network resources needed by the users are locally available. The functions of the access layer are:

- Use of access control and policies
- Creation of separate collision domains
- Workgroup connectivity to the distribution layer

Recap

- A topology defines the physical layout, virtual shape or structure of the network.
- Nodes can be arranged as a mesh, star, bus, or ring topology.
- Bus topology has a single backbone cable and it is very easy to install. Failure in single node affects the whole network.
- Ring topology is arranged in a ring fashion and each node is connected to its neighbouring nodes. It is easier to install but slower than the star topology.
- Star topology has nodes connected to a central controller. Fault isolation is easy but cost of implementation is high.
- In the mesh topology, there is a connection between each node to every other node. It provides great speed but cost of installation is high.
- Nodes can be arranged in two type of architectures: client-server and peer-to-peer.
- In client-server, the node which requests for the service acts as the client and the node which provides the service acts as a server.
- In peer-to-peer, nodes can act as both client and server.
- The Cisco hierarchical model has three layers: core layer, distribution layer and access layer.

Test Your Skills

Multiple Choice Questions

1. Multipoint topology is
   a. Bus
   b. Star
   c. Mesh
   d. Ring

2. If a computer on the network shares resources for others to use, it is called _____
   a. Server
   b. Client
   c. Mainframe
   d. Super Computer
3. For large networks, _____ topology is used.
   a. Bus
   b. Star
   c. Ring
   d. Mesh
4. In _____ topology, if a computer's network cable is broken, the whole network goes down.
   a. Bus
   b. Star
   c. Mesh
   d. Hybrid
5. Which out of the following topologies is the most reliable?
   a. Ring
   b. Star
   c. Bus
   d. None of the above
6. In a network with 25 computers, which one of the following topologies would require the least cabling?
   a. Mesh topology
   b. Star topology
   c. Bus topology
   d. Ring topology